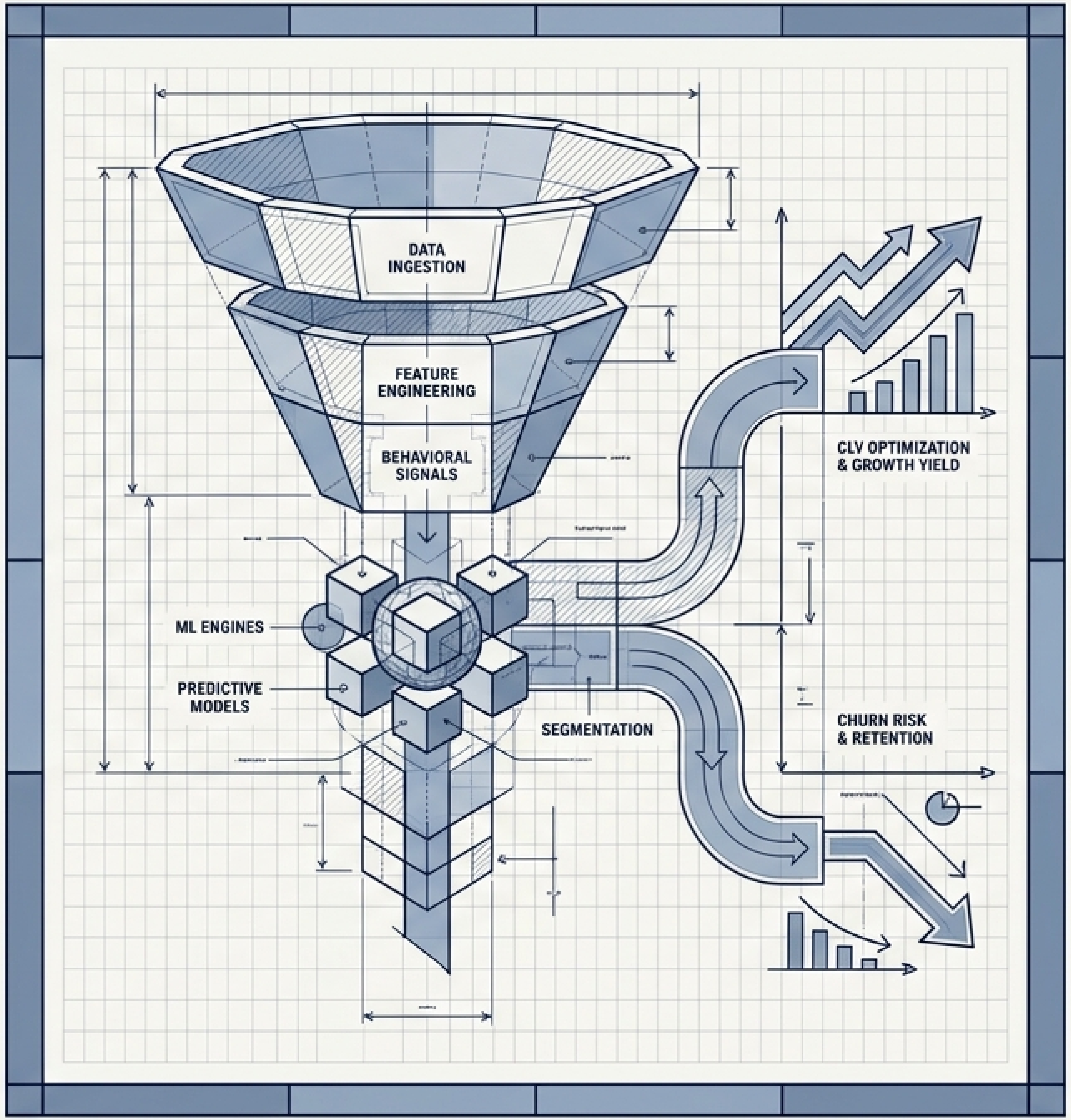
CUSTOMER LIFETIME VALUE & CHURN RISK ANALYSIS

Moving from Reactive Tactics to Predictive CRM

An end-to-end machine learning architecture for forecasting 90-day retention and financial yield



The E-Commerce Retention Dilemma

Traditional / Reactive

- High Customer Acquisition Cost (CAC)
- Recognizing lost customers too late
- Universal flat discount promos
- Margin erosion on guaranteed buyers

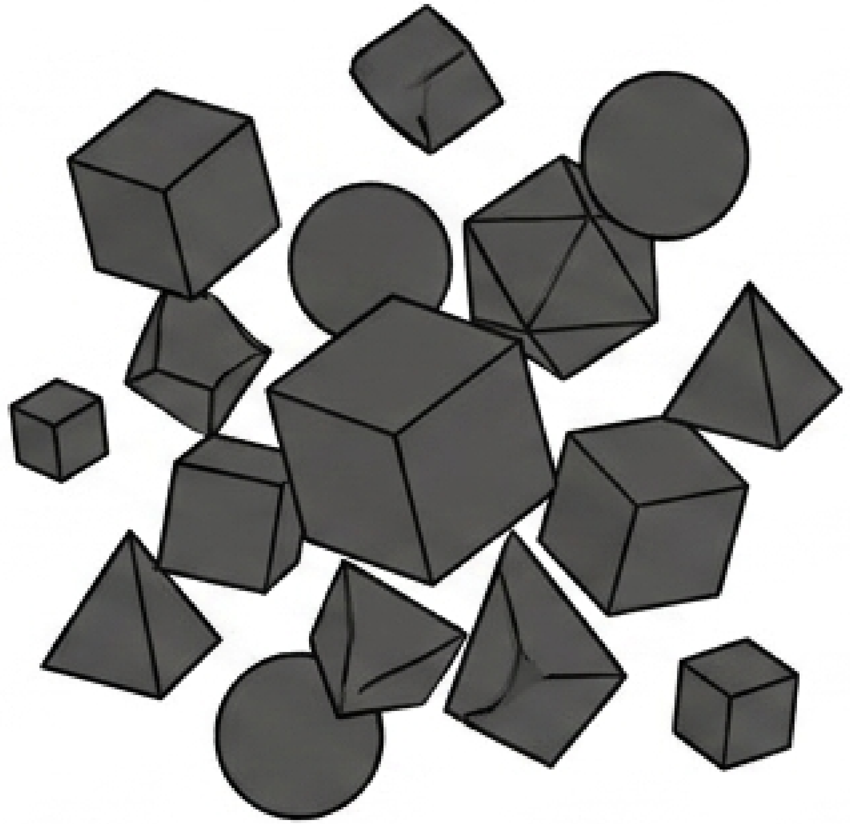
Predictive / Proactive

- Dynamic P(Churn) forecasting
- Targeted high-value win-backs
- Conditional spend estimates
- Budget preservation through automated suppression

In non-contractual retail, a customer doesn't cancel—they just fade away. We must predict the fade.

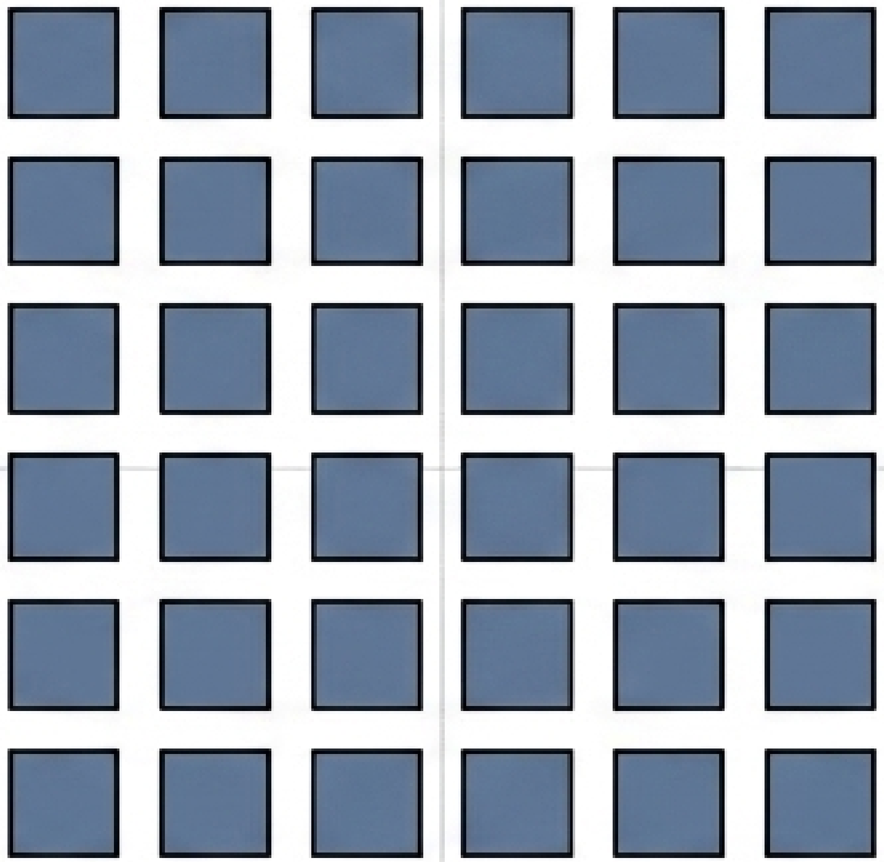
The Data Refinery Pipeline

Stage 1: Raw Ingestion



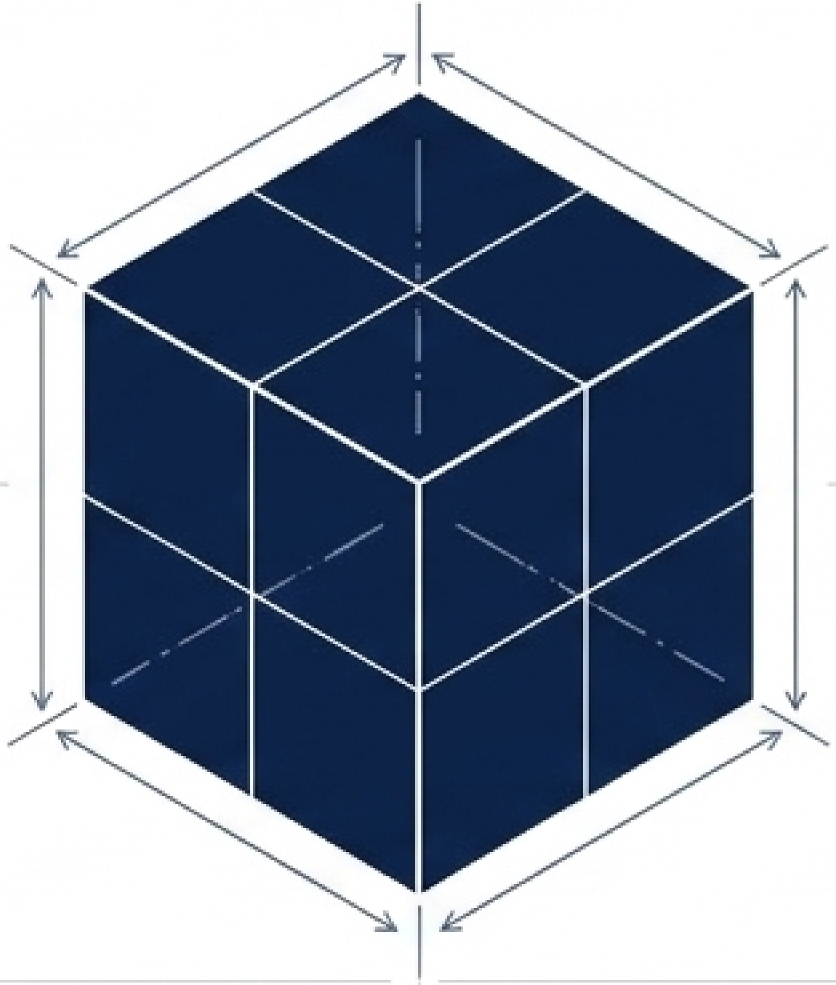
1.06M raw multi-year transactional rows (2009-2011).

Stage 2: Cleansing & Schema



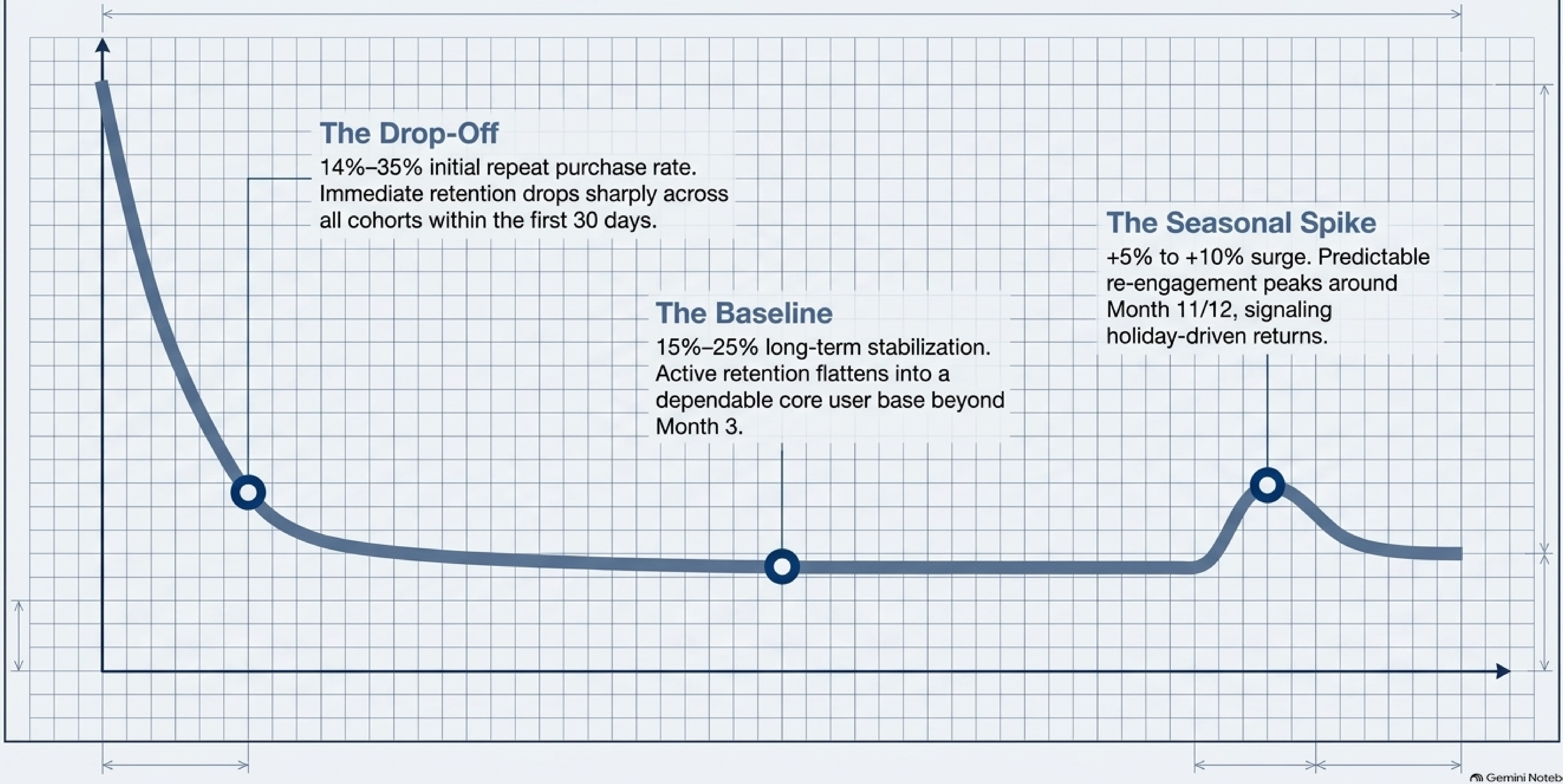
Filtering anomalies, returns, and missing IDs to map 805,549 valid sales to an optimized SQL Star Schema.

Stage 3: Cohort & Features

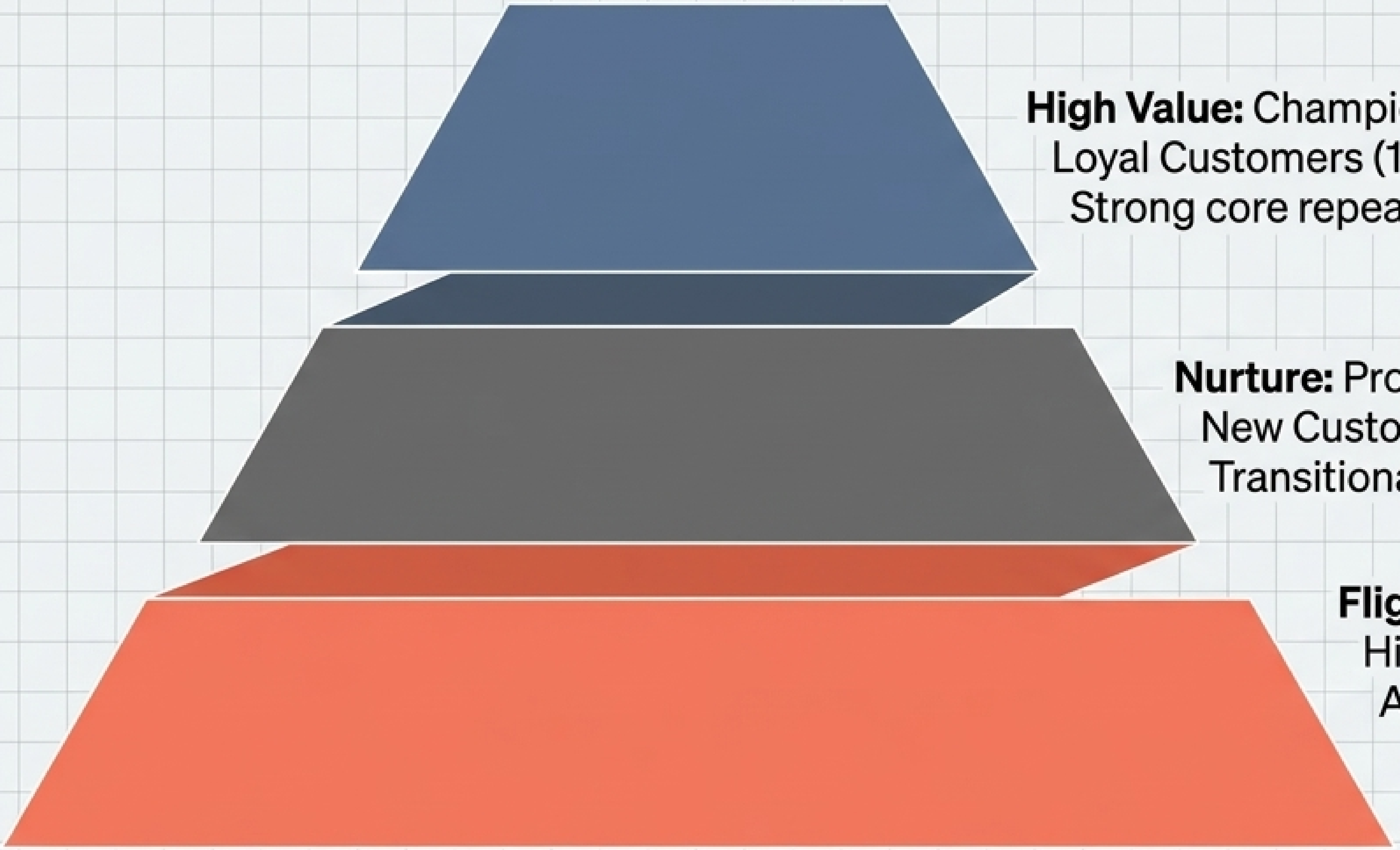


Distilling down to an engineered behavioral feature store of exactly 5,281 unique active customers.

Baseline Behavioral Dynamics



The Customer Landscape: RFM Persona Mapping



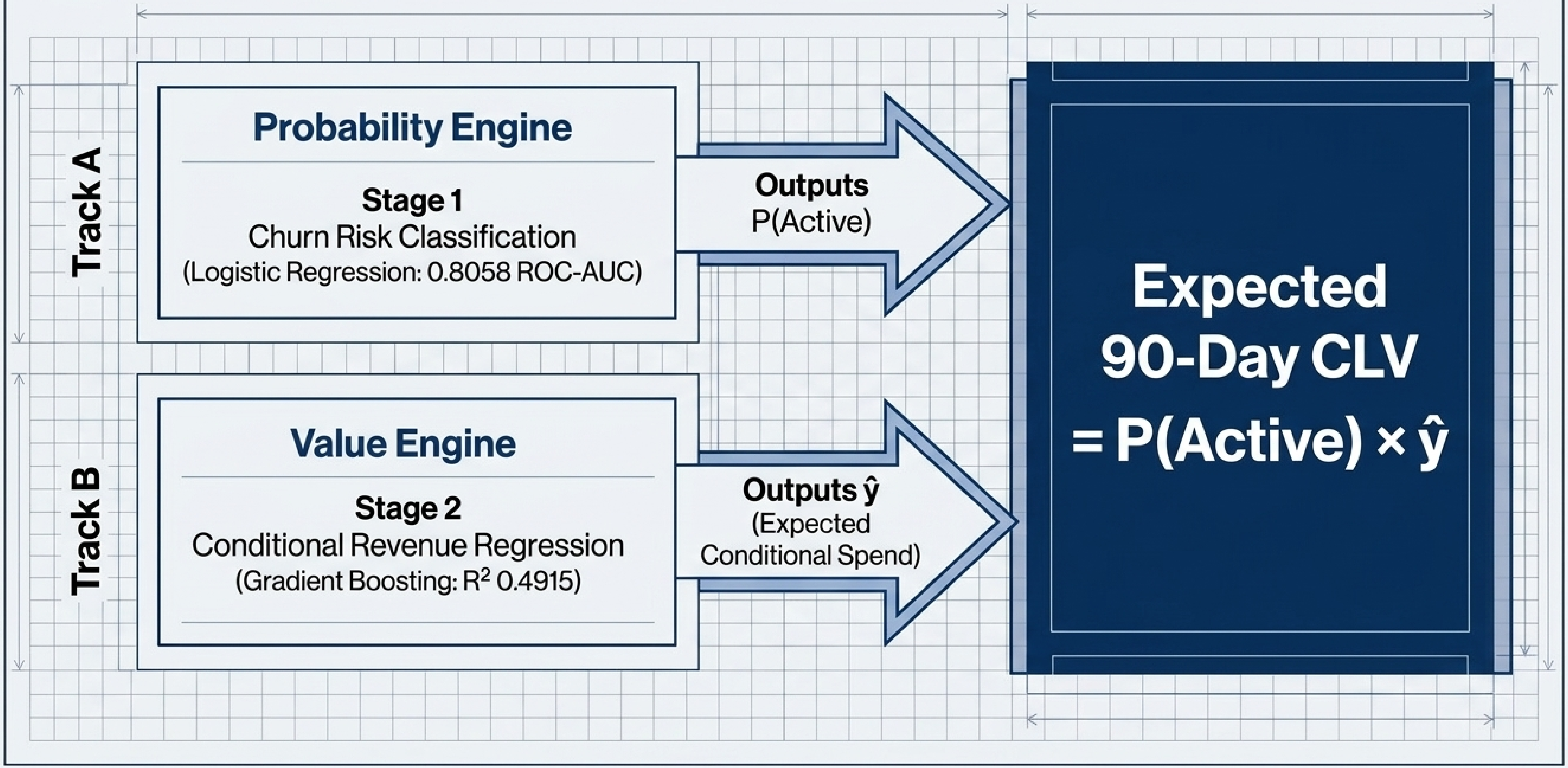
High Value: Champions (26.3%),
Loyal Customers (19.0%).
Strong core repeat purchasing power.

Nurture: Promising / Potential (7.9%),
New Customers (3.2%).
Transitional engagement.

Flight Risk:
Hibernating / Lost (14.9%),
About to Sleep (10.4%),
At Risk (9.6%),
Can’t Lose Them (1.6%).

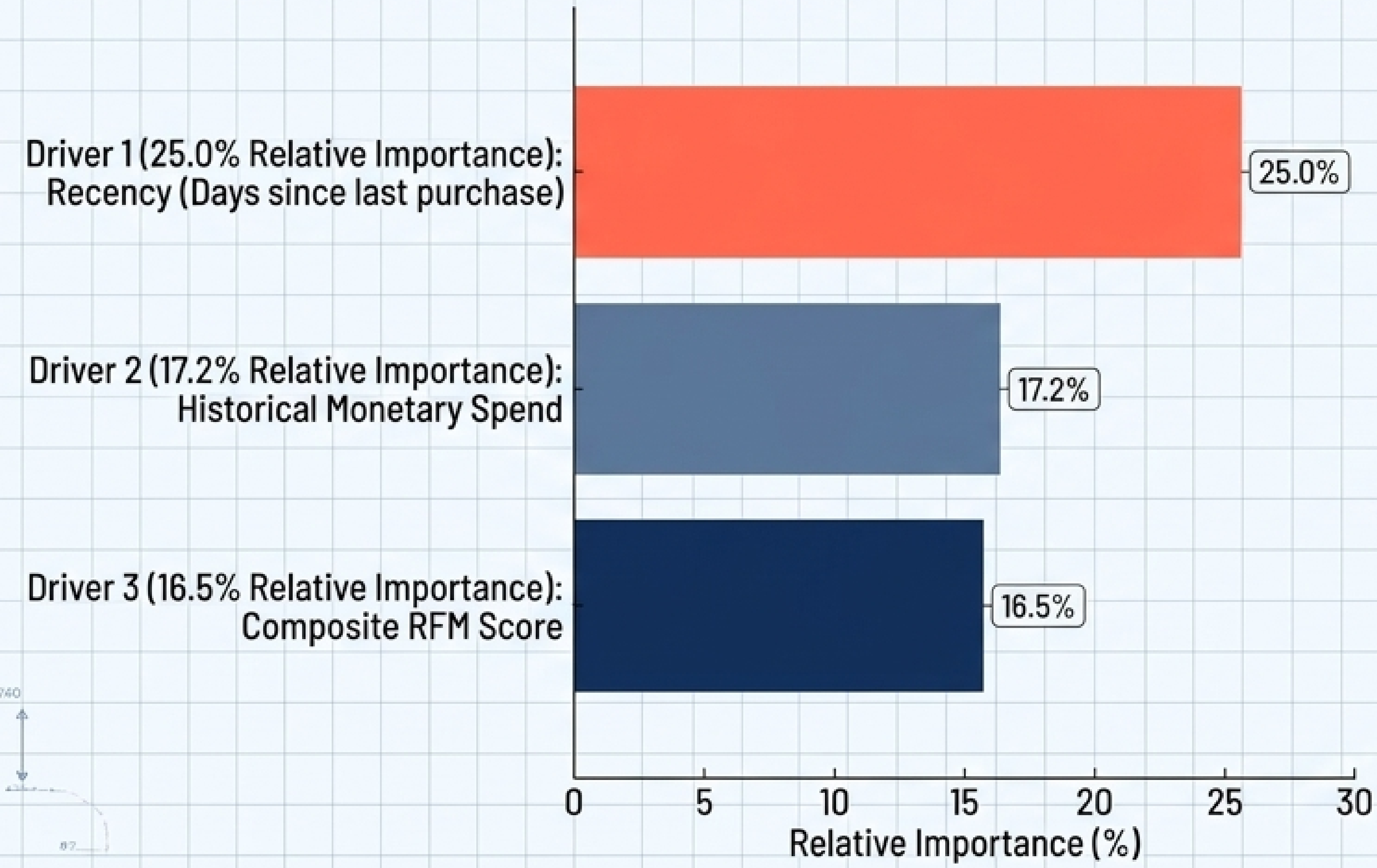
Over 45% of the base forms a highly profitable core, but 36.5% are currently in varying states of critical churn risk.

The Dual-Engine ML Architecture



Editorial Blueprint

Predictive Drivers of Customer Attrition



Empirical validation confirms the model's accuracy:

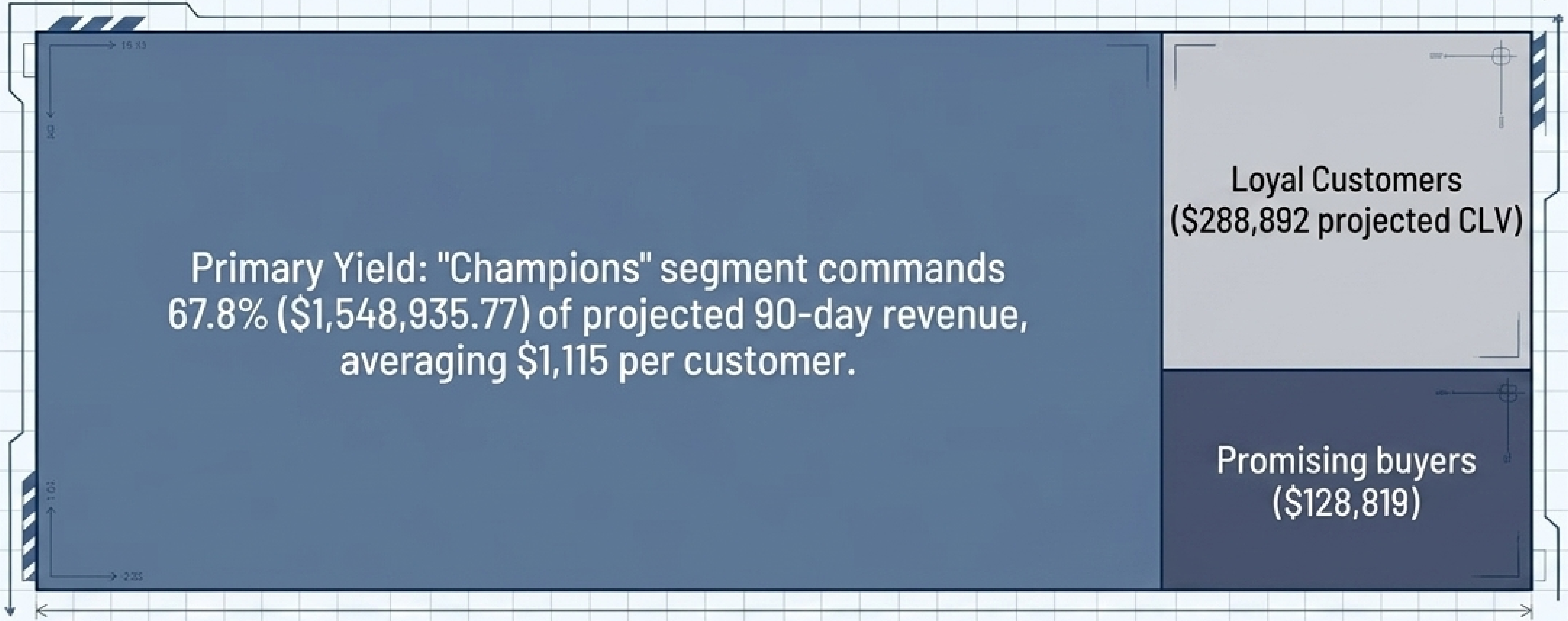
The "Hibernating" persona showed an 89.7% holdout churn rate, while "Champions" showed only 23.3%.

Recency is our ultimate leading indicator of attrition.

Insight Grounding: Based on Holdout Churn Rate by Persona

Editorial Blueprint

Financial Value Concentration



Strategic Directive: Value is fiercely concentrated. Broad-based marketing spreads budget too thin; predictive targeting aligns CRM spend with actual expected yield.

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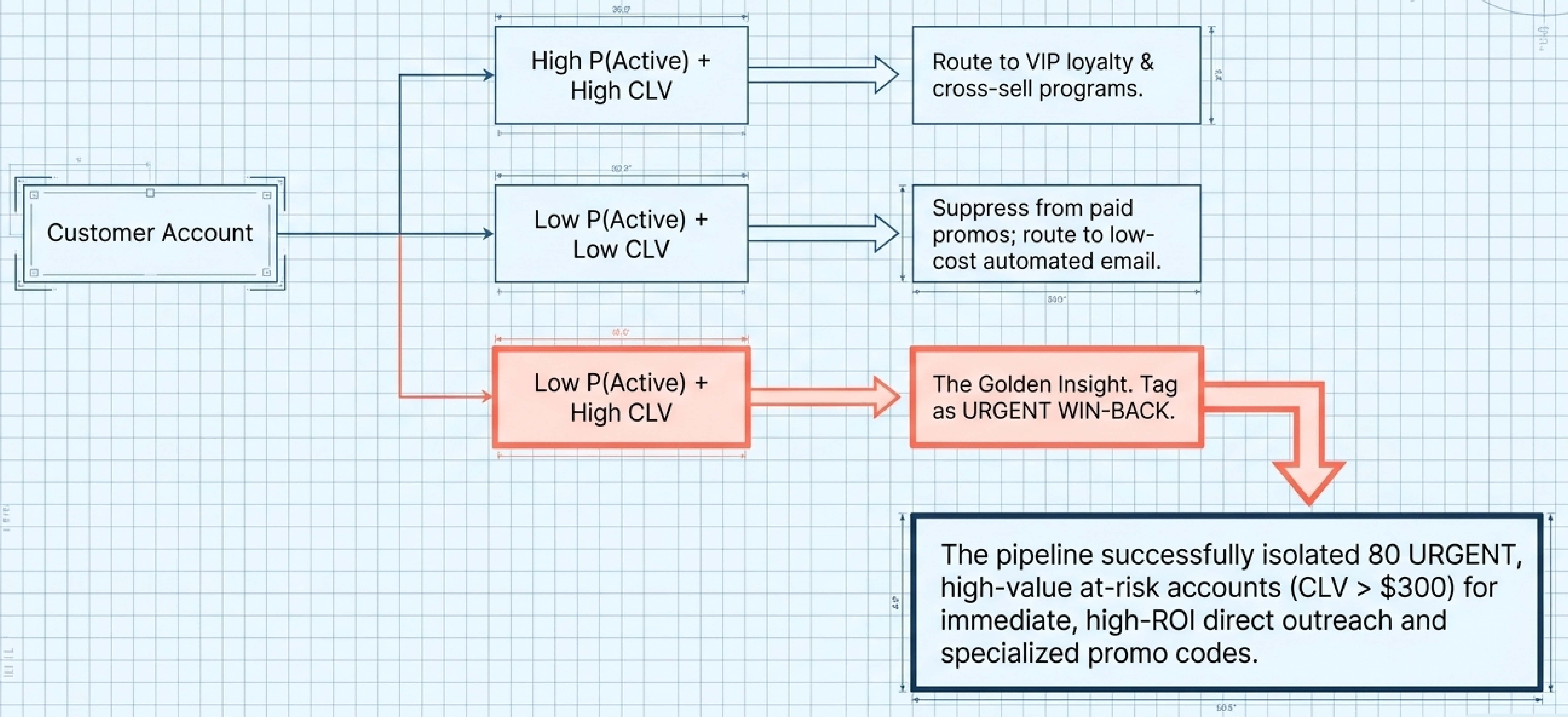
Automated CRM Routing Engine



Editorial Blueprint

Executive Summary & Strategic ROI

Margin Protection

Eliminates indiscriminate discounting by automatically suppressing low-yield and deeply lapsed cohorts.

Proactive Win-Backs

Identifies high-value churn probabilities before the critical re-engagement window permanently closes.

Automated Scalability

The SQLite and dual-stage ML pipeline is fully production-ready for real-time CRM integration.

We have transitioned from viewing historical behavior as a static proxy for the future, to engineering a dynamic, predictive roadmap for ongoing profitability.